

The following are the top operating systems used by hackers and penetration testers. These are under continuous development to make them impenetrable against real-time threats:

- Kali Linux
- Parrot Security OS
- BackBox Linux
- Samurai Web Testing Framework
- Deft Linux
- CAINE
- Network Security Toolkit (NST)
- BlackArch Linux
- ArchStrike Linux
- Cyborg Hawk
- Fedora Security Spin
- BugTraQ
- Node Zero
- Weakerthan
- Dracos Linux

8. Open Source and virtual reality, augmented reality and mixed reality

Recent years have seen an increased interest in the implementation and usage of virtual reality, augmented reality and even mixed reality -- particularly in the areas of healthcare applications, military, fashion, sports, construction, media, telecommunications, films and entertainment, engineering and education. However, higher adoption is limited due to the high costs of developing such software, the lack of technical skills and the difficulties in the implementation environment. To overcome these hurdles, open source has come to the rescue.

Open source has deep roots in this domain, and these will grow deeper and more advanced in the coming few years -- in terms of live implementations, SDKs and software. With the implementation of open source, a new reality called XR or Extended Reality has evolved, which is pretty advanced and has many live implementations.

Virtual reality, augmented reality and mixed reality sales are expected to touch US\$ 2.8 billion by 2018. Various companies like Facebook, Google, Microsoft, Magic Leap, HTC, Samsung, WorldViz, Unity, Snap and FirstHand Technologies have already started adopting open source based technologies for bringing out hardware and software cum SDKs supporting VR, AR and MR.

The Open Source Virtual Reality (OSVR) Consortium and Xilinx are also working on coming out with various VR hardware and software for end users.

The following are various toolkits, platforms and SDKs for VR, AT and MR:

- OSVR - Open Source Virtual Reality for gaming
- ARToolkit - An open source AR SDK
- Apertus VR - An open source VR and AR engine
- OpenSpace3D - An open source platform for 3D environments

- Holokit - An open source MR kit
- Mixare - An open source AR engine
- High Fidelity - An open source shared VR app
- OpenMask - Open source middleware for VR
- CalVR - An open source VR software framework

10. Open source and cloud computing

Cloud computing is one of the most significant technologies to have emerged in the last couple of years and perhaps, even the next couple of years, resulting in billions of dollars in investments. The cloud computing industry is expected to touch US\$ 241 billion by 2020. Common open source cloud applications are CloudStack, OpenNebula or OpenERP servers.

Innovations in cloud computing have led to remarkable trends like the increasing adoption of the public cloud; the adoption of container technologies like Docker, Kubernetes, Apache Mesos, LXD, etc; and the adoption of DevOps in application delivery and open source tooling by companies.

Cloud computing and open source share common goals like minimising costs by not paying licence fees. It is generally accepted that without open source software, cloud computing would not have been able to grow as rapidly. Today most cloud computing vendors use open source software to develop their systems. Examples include Red Hat for cloud operating systems and infrastructure, Eucalyptus and OpenStack for Infrastructure-as-a-Service implementation, Cloudera for the open source Hadoop software framework, OpenNebula for the open source VM, the Xen Supervisor for server virtualisation management, Cloud Foundry and OpenShift for an open Platform-as-a-Service, etc.

The following are some useful tools for cloud computing.

- **OpenStack:** This facilitates data centres to offer the combined resources of computing, storage, networking and a GUI via a dashboard for effective management.
- **CloudStack:** This deploys and manages large networks of virtual machines.
- **Eucalyptus:** This facilitates easy migration of apps and data; it creates private and hybrid cloud environments.
- **openQRM:** Enables the building of private, public and the hybrid IaaS.
- **OpenShift:** Enables easy management of cloud based app development.

Other open source cloud computing simulators include CloudSim, CloudAnalyst, iCanCloud, GreenCloud, CloudSched, etc. 

By: Dr Anand Nayyar

The author is a professor at the Duy Tan University in Vietnam. He loves to work on open source technologies, embedded systems, sensor networks, etc. He can be reached at anand_nayyar@yahoo.co.in. YouTube channel: Gyaan with Anand Nayyar at www.youtube.com/anandnayyar.